Dirichlet's Unit Theorem

A concise, proof-first exposition

Based on Chapter 6 of Robert B. Ash, *A Course in Algebraic Number Theory.*
Organized for a roughly 30-minute mathematical presentation.

Let $K/\mathbb{Q}$ be a number field of degree n, and let $\mathcal{O}_K$ be its ring of integers. There are r_1 embeddings $K \hookrightarrow \mathbb{R}$ and $2r_2$ nonreal embeddings $K \hookrightarrow \mathbb{C}$, the latter occurring in conjugate pairs. Thus $n = r_1 + 2r_2$. Write $\mu(K)$ for the roots of unity in K and $\mathcal{O}_K^\times$ for its group of units.

Theorem 1 (Dirichlet's Unit Theorem)**.** *The roots of unity $\mu(K)$ form a finite cyclic group, and*

$$\boxed{\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r_1+r_2-1}.}$$

In particular, for $r = r_1 + r_2 - 1$, there are units $\varepsilon_1, \dots, \varepsilon_r$ such that every unit is uniquely $\zeta \varepsilon_1^{m_1} \cdots \varepsilon_r^{m_r}$, with $\zeta \in \mu(K)$ and $m_i \in \mathbb{Z}$.

We prove the theorem in three stages: identify the kernel of a logarithmic homomorphism, prove its image is a free abelian group, and use Minkowski's theorem to determine the rank.

1. Logarithmic embedding and its kernel

Choose all r_1 real embeddings and one from each of the r_2 complex-conjugate pairs, denoted $\sigma_1, \dots, \sigma_{r_1+r_2}$. Define

$$\lambda : K^\times \longrightarrow \mathbb{R}^{r_1+r_2}, \qquad \lambda(x) = \big(\log|\sigma_1(x)|, \dots, \log|\sigma_{r_1+r_2}(x)|\big).$$

Since embeddings preserve products, $\lambda(xy) = \lambda(x) + \lambda(y)$.

Lemma 1 (Norm criterion; Ash 6.1.1)**.** *For $x \in \mathcal{O}_K$, we have $x \in \mathcal{O}_K^\times$ if and only if $N_{K/\mathbb{Q}}(x) = \pm 1$.*

Proof. If x is a unit, $N_{K/\mathbb{Q}}(x) N_{K/\mathbb{Q}}(x^{-1}) = 1$; both factors are integers, so $N_{K/\mathbb{Q}}(x) = \pm 1$. Conversely, if $N_{K/\mathbb{Q}}(x) = \pm 1$, the characteristic polynomial of x is monic with integer coefficients and constant term ± 1. By Cayley–Hamilton,

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x \pm 1 = 0,$$

so x times an element of $\mathcal{O}_K$ equals ∓ 1. Hence $x^{-1} \in \mathcal{O}_K$. □

For every $u \in \mathcal{O}_K^\times$, the norm formula and Lemma 1 give

$$1 = |N_{K/\mathbb{Q}}(u)| = \prod_{i=1}^{r_1} |\sigma_i(u)| \prod_{j=r_1+1}^{r_1+r_2} |\sigma_j(u)|^2.$$

Taking logarithms shows that $\lambda(\mathcal{O}_K^\times)$ lies in the hyperplane

$$W = \left\{ y \in \mathbb{R}^{r_1+r_2} : \sum_{i=1}^{r_1} y_i + 2 \sum_{j=r_1+1}^{r_1+r_2} y_j = 0 \right\}, \qquad \dim W = r_1 + r_2 - 1. \tag{1}$$

Lemma 2 (Bounded inverse images; Ash 6.1.3)**.** *For every bounded $C \subset \mathbb{R}^{r_1+r_2}$, the set $\{u \in \mathcal{O}_K^\times : \lambda(u) \in C\}$ is finite.*

Proof. Bounded logarithms imply bounded absolute values for all n conjugates of u (including the omitted conjugate partners). The coefficients of its characteristic polynomial are elementary symmetric functions of these conjugates, so they are bounded. They are also integers because u is integral. Hence only finitely many such polynomials occur, and each has finitely many roots. □

Lemma 3 (Finite subgroups of fields; Ash 6.1.5)**.** *Every finite subgroup G of a field's multiplicative group is cyclic and consists of roots of unity.*

Proof. As a finite abelian group, G has an element g whose order m equals its exponent (the least common multiple of all element orders). Every $x \in G$ satisfies $x^m = 1$. As $X^m - 1$ has at most m roots in a field, $|G| \le m$; but g already generates m distinct elements. Thus $G = \langle g \rangle$. □

Proposition 1 (Kernel; Ash 6.1.4–6.1.6)**.** $\ker(\lambda|_{\mathcal{O}_K^\times}) = \mu(K)$*, which is finite cyclic.*

Proof. Lemma 2 with $C = \{0\}$ makes the kernel finite, so Lemma 3 shows it consists of roots of unity. Conversely, if $\zeta^m = 1$, then $\zeta, \zeta^{-1} \in \mathcal{O}_K$ and $|\sigma_i(\zeta)|^m = 1$ for every i. Thus $\lambda(\zeta) = 0$. □

2. The image is a free abelian group

Put $H = \lambda(\mathcal{O}_K^\times) \subseteq W$. By Lemma 2, every bounded subset of W meets H in finitely many points: H is a discrete subgroup.

Lemma 4 (Discrete subgroup lemma; Ash 6.1.7 and exercises)**.** *If an additive subgroup $H \subseteq \mathbb{R}^m$ meets each bounded set in finitely many points, then $H \cong \mathbb{Z}^s$, where $s = \dim_{\mathbb{R}} \operatorname{span}_{\mathbb{R}} H \le m$.*

Proof. Choose $e_1, \dots, e_s \in H$ that are maximally independent over $\mathbb{R}$. Every $h \in H$ is $h = \sum a_i e_i$. The bounded parallelepiped $P = \{\sum t_i e_i : 0 \le t_i \le 1\}$ contains only finitely many points of H. Reducing coefficients modulo $\mathbb{Z}$ shows that H is generated by $e_1, \dots, e_s$ and the finite set $H \cap P$.

Moreover, for any $h = \sum a_i e_i$ and each positive integer j, $jh - \sum \lfloor j a_i \rfloor e_i \in H \cap P$. Two such elements coincide, forcing $(j-k)a_i \in \mathbb{Z}$ for some $j \ne k$ and every i. Thus the a_i are rational. Taking a common denominator d for a finite generating set gives $dH \subseteq \bigoplus_{i=1}^s \mathbb{Z} e_i$. A subgroup of $\mathbb{Z}^s$ is free, and the e_i ensure its rank is exactly s. □

Apply Lemma 4 to $H \subseteq W$ and the first isomorphism theorem:

$$H \cong \mathbb{Z}^s, \qquad s \le \dim W = r_1 + r_2 - 1, \qquad \mathcal{O}_K^\times / \mu(K) \cong H.$$

Choose units whose logarithmic images form a $\mathbb{Z}$-basis of H. Their lifts split the quotient, giving

$$\boxed{\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^s, \qquad s \le r_1 + r_2 - 1.} \tag{2}$$

It remains to prove that equality holds.

3. Minkowski's theorem gives the exact rank

Write $r = r_1 + r_2 - 1$ and $V = \operatorname{span}_{\mathbb{R}} H \subseteq W$. If $r = 0$, equation (2) already proves the theorem. Assume $r > 0$. It suffices to prove $V = W$.

Step 1: Construct integral elements with controlled conjugates. The canonical embedding $\sigma : \mathcal{O}_K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n$ is a lattice of covolume $2^{-r_2}\sqrt{|D_K|}$, where D_K is the field discriminant. Choose a fixed

$$b > \max\left\{1, \left(\frac{2}{\pi}\right)^{r_2} \sqrt{|D_K|}\right\}.$$

Take arbitrary $a_1, \ldots, a_r > 0$; choose the remaining $a_{r+1} > 0$ so that

$$\prod_{i=1}^{r_1} a_i \prod_{j=r_1+1}^{r_1+r_2} a_j^2 = b. \tag{3}$$

Consider the symmetric convex body

$$S = \{(y, z) \in \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} : |y_i| \le a_i,\ |z_j| \le a_{r_1+j}\}.$$

Its volume is $2^{r_1}\pi^{r_2} b > 2^n \operatorname{covol}(\sigma(\mathcal{O}_K))$. Minkowski's convex body theorem gives $0 \neq x \in \mathcal{O}_K$ with

$$|\sigma_i(x)| \le a_i \qquad (1 \le i \le r_1 + r_2). \tag{4}$$

Step 2: Uniform logarithmic control. The norm is a nonzero integer, so equations (3)–(4) give $1 \le |N_{K/\mathbb{Q}}(x)| \le b$. For a fixed embedding, bound the product of all *other* conjugate absolute values by b/a_i (counting each complex-conjugate partner). Then

$$\frac{a_i}{b} \le |\sigma_i(x)| \le a_i, \qquad 0 \le \log a_i - \log|\sigma_i(x)| \le \log b. \tag{5}$$

Lemma 5 (Bounded ideal norms; used in Ash 6.2.1)**.** *For any $M > 0$, only finitely many nonzero integral ideals $I \subseteq \mathcal{O}_K$ satisfy $N(I) \le M$.*

Proof. If $N(I) = m$, the additive group $\mathcal{O}_K/I$ has order m, so $m \in I$ and $(m) \subseteq I$. Ideals $I \supseteq (m)$ correspond to ideals of the finite ring $\mathcal{O}_K/(m)$, of which there are finitely many. There are only finitely many integers $1 \le m \le M$. □

Step 3: Force a unit in every linear direction. Let $f : W \to \mathbb{R}$ be a nonzero linear functional. Since the last coordinate on W is determined by the first r, write $f(y) = \sum_{i=1}^{r} c_i y_i$ on W, with some $c_i \neq 0$. Use the same formula to extend f to a functional F on $\mathbb{R}^{r+1}$; we need this extension for x that may not be units.

By equation (5), every element x produced in Step 1 satisfies

$$\left| F(\lambda(x)) - \sum_{i=1}^{r} c_i \log a_i \right| \le \left(\sum_{i=1}^{r} |c_i|\right) \log b =: C. \tag{6}$$

Choose $t > C$. For each integer $h \ge 1$, select the free parameters $a_i > 0$ so that $\sum_i c_i \log a_i = 2th$; this is possible because some $c_i \neq 0$. The constructed nonzero algebraic integer x_h satisfies

$$(2h-1)t < F(\lambda(x_h)) < (2h+1)t, \qquad 1 \le |N_{K/\mathbb{Q}}(x_h)| \le b.$$

The displayed intervals are pairwise disjoint, so the values $F(\lambda(x_h))$ are distinct. Yet the principal ideals (x_h) have bounded norm $N((x_h)) = |N_{K/\mathbb{Q}}(x_h)| \le b$. By Lemma 5, two coincide: $(x_h) = (x_k)$ for some $h \neq k$. Hence $x_h = ux_k$ for some $u \in \mathcal{O}_K^\times$. Taking logarithms,

$$f(\lambda(u)) = F(\lambda(x_h)) - F(\lambda(x_k)) \neq 0.$$

Therefore no nonzero linear functional on W vanishes on V. Hence $V = W$, so $s = \dim W = r_1 + r_2 - 1$. Substituting into equation (2) proves Theorem 1. □

4. Applications: quadratic fields

Imaginary quadratic fields. Let $K = \mathbb{Q}(\sqrt{m})$ with $m < 0$ square-free. Here $(r_1, r_2) = (0, 1)$, so Dirichlet gives $\mathcal{O}_K^\times = \mu(K)$: all units are roots of unity. If $m \not\equiv 1 \pmod 4$, write $u = a + b\sqrt{m}$, and the norm criterion becomes $a^2 + |m|b^2 = 1$. If $m \equiv 1 \pmod 4$, write $u = (a + b\sqrt{m})/2$ with $a \equiv b \pmod 2$, and obtain $a^2 + |m|b^2 = 4$. Solving these equations yields

$$\boxed{\mathcal{O}_K^\times = \begin{cases} \{\pm 1, \pm i\}, & m = -1, \\ \{\pm 1, \pm\omega, \pm\omega^2\}, & m = -3,\ \omega = (-1 + \sqrt{-3})/2, \\ \{\pm 1\}, & \text{otherwise.} \end{cases}}$$

Real quadratic fields. If $m > 1$ is square-free, then $(r_1, r_2) = (2, 0)$ and $\mu(K) = \{\pm 1\}$. Thus

$$\boxed{\mathcal{O}_K^\times = \{\pm\varepsilon^k : k \in \mathbb{Z}\},}$$

where $\varepsilon > 1$ is the *fundamental unit*, the smallest unit greater than 1. It is found by the Pell-type norm conditions

$$\begin{aligned} m \not\equiv 1 \pmod 4 &: \quad u = a + b\sqrt{m}, \quad a^2 - mb^2 = \pm 1, \\ m \equiv 1 \pmod 4 &: \quad u = (a + b\sqrt{m})/2, \quad a^2 - mb^2 = \pm 4 \quad (a \equiv b \pmod 2). \end{aligned}$$

In each case, the least positive b giving a solution with $a > 0$ produces ε; continued fractions give a faster algorithm. For instance, $\varepsilon_{\mathbb{Q}(\sqrt{2})} = 1 + \sqrt{2}$ and $\varepsilon_{\mathbb{Q}(\sqrt{5})} = (1 + \sqrt{5})/2$.

A higher-degree example. For an odd prime p, the cyclotomic field $\mathbb{Q}(\zeta_p)$ has $r_1 = 0$ and $r_2 = (p-1)/2$, so its unit group has free rank $(p - 3)/2$.

Reference. R. B. Ash, *A Course in Algebraic Number Theory*, Chapter 6, Sections 6.1–6.3. The supporting lattice and ideal-norm facts are developed in Chapters 4–5.